

Practical

Subject: Mobile Application Development

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Roll No: 3

Semester: - 5

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Practical 5A : Larger of Two Numbers

Aim: To develop an Android application that accepts two numbers as input from the user and displays the larger of the two numbers using conditional logic.

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:orientation="vertical"

    android:gravity="center"

    android:padding="20dp">

    <EditText

        android:id="@+id/num1"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:hint="Enter First Number"

        android:inputType="numberSigned"/>

    <EditText

        android:id="@+id/num2"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:hint="Enter Second Number"
```

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```
android:inputType="numberSigned"/>
```

```
<Button
```

```
    android:id="@+id/btn"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"
```

```
    android:text="Find Larger"/>
```

```
<TextView
```

```
    android:id="@+id/result"
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:textSize="20sp"/>
```

```
</LinearLayout>
```

MainActivity.java:

```
package com.example.practical5a;
```

```
import android.os.Bundle;
```

```
import android.widget.*;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    EditText num1, num2;
```

```
    Button btn;
```

```
    TextView result;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);
```

```
        num1 = findViewById(R.id.num1);
```

```
        num2 = findViewById(R.id.num2);
```

```
        btn = findViewById(R.id.btn);
```

```
        result = findViewById(R.id.result);
```

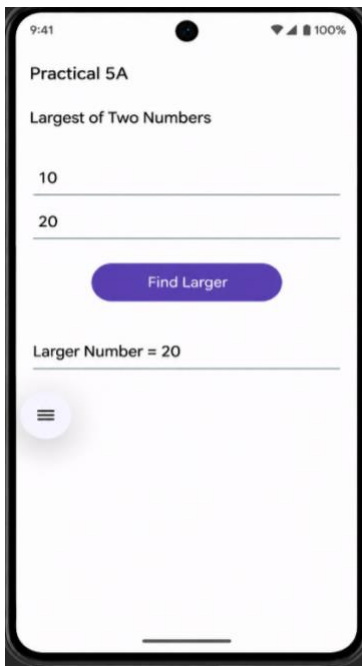
```
        btn.setOnClickListener(v -> {
```

```
            int a = Integer.parseInt(num1.getText().toString());
```

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```
int b = Integer.parseInt(num2.getText().toString());  
if (a > b)  
    result.setText("Larger Number = " + a);  
else  
    result.setText("Larger Number = " + b);  
});  
}  
}
```

Expected Output:



Practical 5B : Positive or Negative

Aim: To develop an Android application that accepts a number as input and determines whether it is positive or negative.

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">
    <EditText
        android:id="@+id/num"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Number"
        android:inputType="numberSigned"/>
    <Button
        android:id="@+id/btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Check"/>
    <TextView
        android:id="@+id/result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.practical5b;

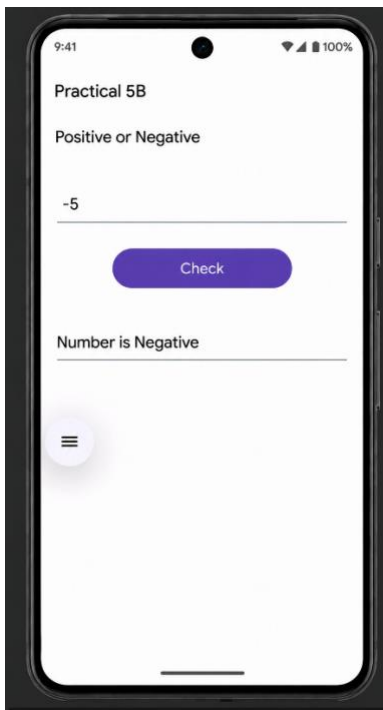
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText num;
    Button btn;
    TextView result;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        num = findViewById(R.id.num);
```

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```
btn = findViewById(R.id.btn);
result = findViewById(R.id.result);
btn.setOnClickListener(v -> {
    int n = Integer.parseInt(num.getText().toString());
    if (n >= 0)
        result.setText("Positive Number");
    else
        result.setText("Negative Number");
});
}
```

Expected Output:



Practical 5C : Vowel or Consonant

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Aim: To develop an Android application that accepts a character as input and determines whether it is a vowel or a consonant.

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="30dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Vowel or Consonant"
        android:textSize="25sp"
        android:textStyle="bold"/>

    <EditText
        android:id="@+id/character"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter a Character"
        android:maxLength="1"/>

    <Button
        android:id="@+id/checkButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Check"/>

    <TextView
        android:id="@+id/result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp"/>
</LinearLayout>
```

MainActivity.java:

```
package com.example.vowel;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
```

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EditText character;
Button checkButton;
TextView result;

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    character = findViewById(R.id.character);
    checkButton = findViewById(R.id.checkButton);
    result = findViewById(R.id.result);

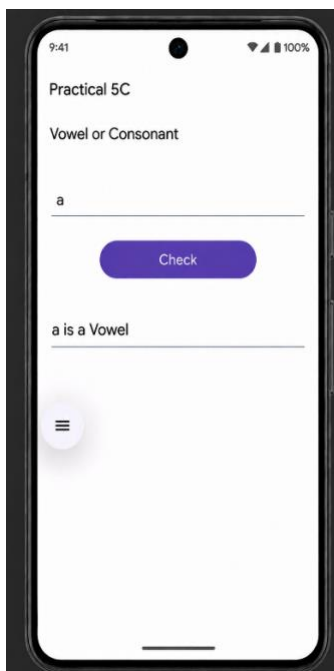
    checkButton.setOnClickListener(v -> {

        char ch = character.getText().toString().charAt(0);

        if(ch=='a'||ch=='e'||ch=='i'||ch=='o'||ch=='u'
        || ch=='A'||ch=='E'||ch=='I'||ch=='O'||ch=='U')
            result.setText("It is a Vowel");
        else
            result.setText("It is a Consonant");

    });
}
```

Expected Output:



Practical 5D : Reverse Number

Aim: To develop an Android application that accepts a number as input and displays its reverse using loop-based digit manipulation.

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/num"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Number"
        android:inputType="number"/>
```

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```
<Button
    android:id="@+id/btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Reverse"/>

<TextView
    android:id="@+id/result"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>
</LinearLayout>
```

MainActivity.java:

```
package com.example.reverse;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText num;
    Button btn;
    TextView result;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        num = findViewById(R.id.num);
        btn = findViewById(R.id.btn);
        result = findViewById(R.id.result);

        btn.setOnClickListener(v -> {

            int n = Integer.parseInt(num.getText().toString());
            int rev = 0;

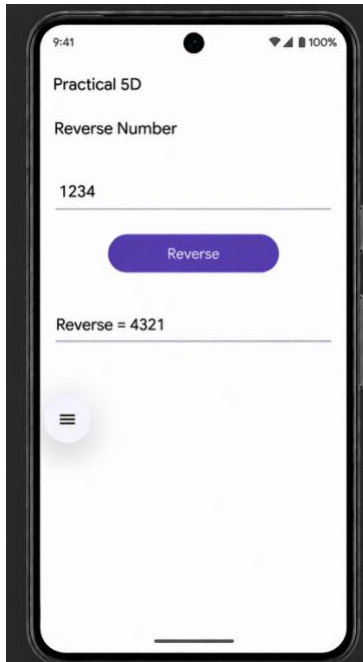
            while(n != 0){
                rev = rev * 10 + n % 10;
                n = n / 10;
            }

            result.setText("Reverse = " + rev);

        });
    }
}
```

}

Expected Output:



Practical 6A : Hello World

Aim: To create a basic Android application that displays a "Hello World" message on the screen using a TextView.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <TextView
        android:id="@+id/tv"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World"
        android:textSize="30sp"
        android:textStyle="bold"/>
</LinearLayout>
```

MainActivity.java

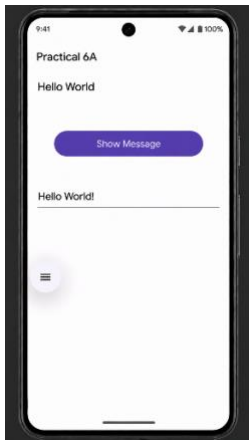
```
package com.example.hello;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Expected Output:



Practical 6B : Toast Message

Aim: To develop an Android application that displays a Toast message on the screen when a button is clicked.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Toast"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.toast;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

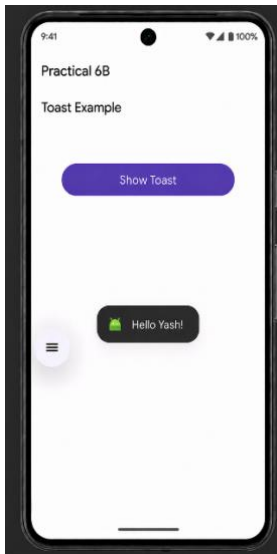
public class MainActivity extends AppCompatActivity {
```

Button button;

@Override

```
protected void onCreate(Bundle savedInstanceState) {  
    super.onCreate(savedInstanceState);  
    setContentView(R.layout.activity_main);  
  
    button = findViewById(R.id.button);  
  
    button.setOnClickListener(v ->  
        Toast.makeText(this, "Hello Yash",  
            Toast.LENGTH_SHORT).show()  
    );  
}
```

Expected Output:



Practical 6C : Login Form

Aim: To develop an Android login application that validates a username and password entered by the user and displays an appropriate Toast message.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="30dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login"
        android:textSize="30sp"
        android:textStyle="bold"/>

    <EditText
        android:id="@+id/username"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"/>

    <EditText
        android:id="@+id/password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"/>
```

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```
<Button
    android:id="@+id/loginButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Login"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.login;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText username, password;
    Button loginButton;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        username = findViewById(R.id.username);
        password = findViewById(R.id.password);
        loginButton = findViewById(R.id.loginButton);

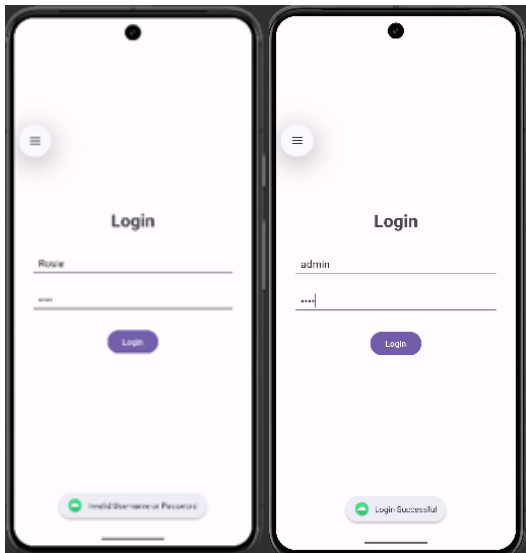
        loginButton.setOnClickListener(v -> {

            String user = username.getText().toString();
            String pass = password.getText().toString();

            if(user.equals("admin") && pass.equals("1234"))
                Toast.makeText(this,"Login Successful",Toast.LENGTH_SHORT).show();
            else
                Toast.makeText(this,"Invalid Username or Password",Toast.LENGTH_SHORT).show();

        });
    }
}
```

Expected Output



Practical 7A : Dialer App

Aim: To develop an Android application that uses an Intent to launch the phone dialer with a user-entered phone number.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/phone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Phone Number"
        android:inputType="phone"/>

    <Button
        android:id="@+id/dial"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Dial"/>
</LinearLayout>
```

MainActivity.java

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```
package com.example.dialer;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

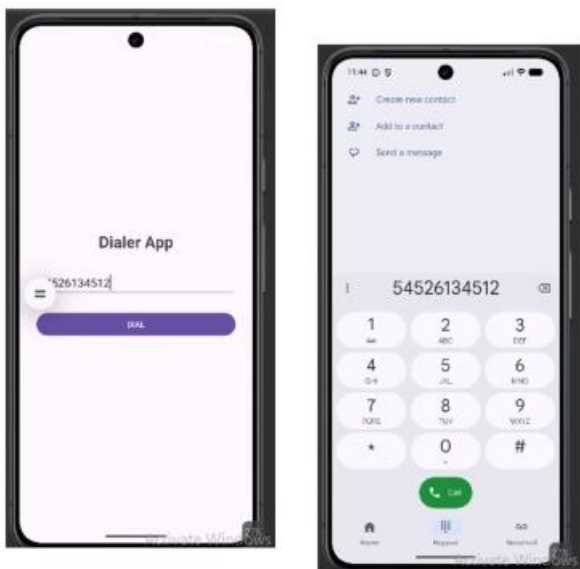
    EditText phone;
    Button dial;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        phone = findViewById(R.id.phone);
        dial = findViewById(R.id.dial);

        dial.setOnClickListener(v -> {
            Intent i = new Intent(Intent.ACTION_DIAL,
                Uri.parse("tel:" + phone.getText().toString()));
            startActivity(i);
        });
    }
}
```

Expected Output:



Practical 7B : Browser App

Aim: To develop an Android application that uses an Intent to open a web browser with a user-entered website URL.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/url"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Website"/>

    <Button
        android:id="@+id/open"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Open"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.browser;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText url;
    Button open;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

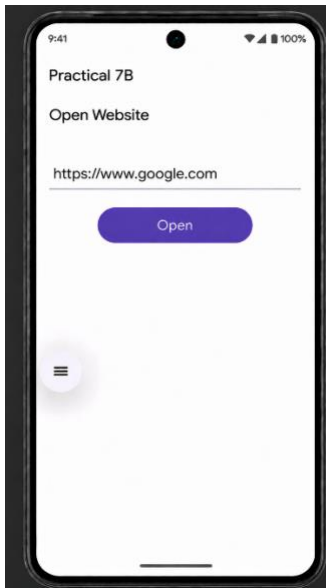
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```
setContentView(R.layout.activity_main);

url = findViewById(R.id.url);
open = findViewById(R.id.open);

open.setOnClickListener(v -> {
    Intent i = new Intent(Intent.ACTION_VIEW,
        Uri.parse("https://" + url.getText().toString()));
    startActivity(i);
});
}
```

Expected Output:



Practical 8A : SQLite Database

Aim: To develop an Android application that stores student details such as roll number, name, and age using an SQLite database.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/roll"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Roll No"/>

    <EditText
        android:id="@+id/name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Student Name"/>

    <EditText
        android:id="@+id/age"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Age"/>

    <Button
        android:id="@+id/save"
        android:layout_width="match_parent"
```

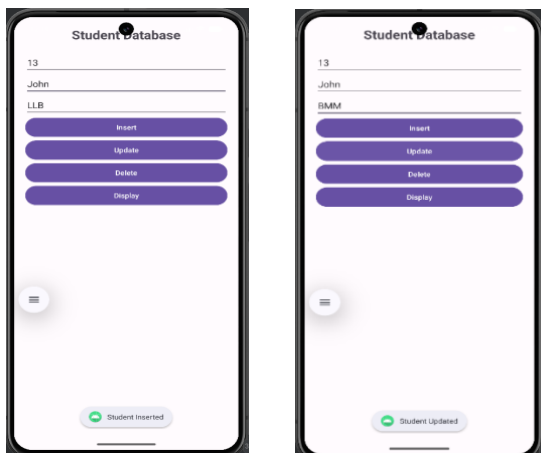
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```
        android:layout_height="wrap_content"  
        android:text="Save"/>  
</LinearLayout>
```

MainActivity.java

```
package com.example.sqlite;  
  
import android.os.Bundle;  
import android.widget.*;  
import androidx.appcompat.app.AppCompatActivity;  
  
public class MainActivity extends AppCompatActivity {  
  
    EditText roll, name, age;  
    Button save;  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_main);  
  
        roll = findViewById(R.id.roll);  
        name = findViewById(R.id.name);  
        age = findViewById(R.id.age);  
        save = findViewById(R.id.save);  
  
        save.setOnClickListener(v ->  
            Toast.makeText(this,  
                "Student Saved Successfully",  
                Toast.LENGTH_SHORT).show()  
        );  
    }  
}
```

Expected Output:



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Student Database

13

John

BMM

Insert

Update

Delete

Display

Roll No: 13
Name: John
Course: BMM

Student Deleted

Student Database

13

John

BMM

Insert

Update

Delete

Display

Roll No: 13
Name: John
Course: BMM

Student Database

13

John

BMM

Insert

Update

Delete

Display

students			
Live updates			
	rollno	name	course
1	13	John	BMM
2	16	Harry	CA

Practical 8B : Intent Navigation

Aim: To develop an Android application that demonstrates navigation between two activities using an Intent.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <Button
        android:id="@+id/next"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Next Activity"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.intent;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button next;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

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```
next = findViewById(R.id.next);

next.setOnClickListener(v ->
    startActivity(new Intent(MainActivity.this,
        SecondActivity.class))
    );
}
```

second_activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Second Activity!"
        android:textSize="22sp"/>

    <Button
        android:id="@+id/back"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Go Back"/>
</LinearLayout>
```

Expected Output:

